Office of Engineering Research Administration
College of Engineering
The Pennsylvania State University
101 Hammond Building
University Park, PA 16802 Phone: (814) 865-3272

Engineering Proposal Transmittal Form

Proposal Submission Date	April 20 2026
Proposal Number *(Please use this number in all correspondence)*	OSP# 260244
Approved by The Pennsylvania State University:	Raquel Andersen-Hubby _____ Raquel Andersen-Hubby On Behalf of: Robin B. Riglin, JD Assistant Vice President for Research and Director, Office of Sponsored Programs
Submitted To:	Argonne National Laboratory 9700 South Cass Avenue, Building 240, Room 1141 Lemont, IL 60439 USA Email: mmin@mcs.anl.gov
The enclosed proposal is submitted for your consideration and is entitled:	***Foundation Models for Automated Design of Energy Flow Systems***
Technical Contact *For information pertaining to the technical portions of the proposal please contact:*	PI - Dr. Elia Merzari Professor Department of Nuclear Engineering The Pennsylvania State University 228 Hallowell Bldg, University Park, PA 16802 Phone: (814) 863-7469 Email: ebm5351@psu.edu
Administrative/Fiscal Contact *For information pertaining to the administrative or financial portions of the proposal; or to request revisions to the proposal please contact:*	Raquel Andersen-Hubby Senior Proposal Generalist The Pennsylvania State University Office of Sponsored Programs 110 Technology Center Building, University Park, PA 16802 Phone: 814-863-1246 / Email: rla5452@psu.edu
Contractual Contact *Contractual negotiations will occur with:*	Office of Sponsored Programs 200 Innovation Blvd., Suite 110 University Park, PA 16802 Phone: 814-865-1372 / Email: osp@psu.edu
Transmittal Notes *These notes contain additional information for the sponsor, or take exception to sponsor provided terms and conditions:*	The Pennsylvania State University employs individuals and accepts students and graduate research students from a multitude of national backgrounds. As an entity, the University is subject to, and works diligently to obey, federal regulations regarding the export of controlled technologies and data. Sponsor, as an independent entity, is individually responsible for ascertaining its compliance with federal export laws and procedures. If Sponsor anticipates disclosure or provision of controlled technology or data to University as part of the proposed sponsored project, Sponsor should inform University, in writing, of the existence of, and information concerning the scope and extent of, such anticipated disclosures or provisions. If this proposal is selected for funding, The Pennsylvania State University shall negotiate terms and conditions of the award which are applicable to non-profit educational institutions.

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April 20, 2026

Misun Min
9700 South Cass Avenue, Building 240, Room 1141
Lemont, IL 60439 USA
Email: mmin@mcs.anl.gov

Subject: Letter of Commitment for Proposal entitled "***Foundation Models for Automated Design of Energy Flow Systems***" [Ref. Penn State OSP# 260244]

To Whom it May Concern:

The Pennsylvania State University through the College of Engineering is pleased to confirm our participation on the subject proposal led by Argonne National Laboratory in response to DOE's Request for Applications, "The Genesis Mission: Transforming Science and Energy with AI" (DE-FOA-0003612). The proposal is entitled "Foundation Models for Automated Design of Energy Flow Systems" with Dr. Elia Merzari serving as the Penn State Principal Investigator. Support is requested for the period of 07/01/2026 – 03/31/2027 and is in the total amount of $70,000. This proposal has been reviewed and approved by the appropriate administrative personnel at Penn State.

Penn State will contribute to the project by helping develop high-fidelity numerical models that will be used to train and benchmark the foundation model. In addition, Penn State will support the development of one or more benchmarks and challenge problems for nuclear fluid flow and heat transfer applications.

If we can be of further assistance, please contact Raquel Andersen-Hubby-Senior Proposal Generalist, via phone: 814-863-1246 or email: rla5452@psu.edu. Technical questions should be directed to the PI, Dr. Elia Merzari, via phone: 814-863-7469 or email: ebm5351@psu.edu.

Sincerely,

Raquel Andersen-Hubby

Raquel Andersen-Hubby-Senior Proposal Generalist
On Behalf of:
Robin B. Riglin
Assistant Vice President for Research and
Director, Office of Sponsored Programs

RESEARCH & RELATED Senior/Key Person Profile

PROFILE - Project Director/Principal Investigator

Prefix: Dr. * First Name: Elia Middle Name:

* Last Name: Merzari Suffix:

Position/Title:

Department:

Organization Name: The Pennsylvania State University

Division:

* Street1: 228 Hallowell Building, 211 N Barnard St

Street2:

* City: University Park County: Centre

* State: PA: Pennsylvania Province:

* Country: USA: UNITED STATES * Zip / Postal Code: 16802-7000

* Phone Number: 814-863-7469 Fax Number:

* E-Mail: ebm5351@psu.edu

Credential, e.g., agency login:

*** Project Role:** PD/PI **Other Project Role Category:**

*** Attach Biographical Sketch** CV_ Merzari.pdf Add Attachment Delete Attachment View Attachment

Attach Current & Pending Support CPOS_EM.pdf Add Attachment Delete Attachment View Attachment

PROFILE - Senior/Key Person 1

Prefix: * First Name: Middle Name:

* Last Name: Suffix:

Position/Title:

Department:

Organization Name:

Division:

* Street1:

Street2:

* City: County:

* State: Province:

* Country: * Zip / Postal Code:

* Phone Number: Fax Number:

* E-Mail:

Credential, e.g., agency login:

*** Project Role:** PD/PI **Other Project Role Category:**

*** Attach Biographical Sketch** Add Attachment Delete Attachment View Attachment

Attach Current & Pending Support Add Attachment Delete Attachment View Attachment

Next Person

ADDITIONAL SENIOR/KEY PERSON PROFILE(S) Add Attachment Delete Attachment View Attachment

Additional Biographical Sketch(es) (Senior/Key Person) Add Attachment Delete Attachment View Attachment

Additional Current and Pending Support(s) Add Attachment Delete Attachment View Attachment

OMB Number: 4040-0001
Expiration Date: 01/31/2029

IDENTIFYING INFORMATION:

NAME: Merzari, Elia

POSITION TITLE: Professor

PRIMARY ORGANIZATION AND LOCATION: Pennyslvania State University, State College, Pennsylvania, United States

Professional Preparation:

ORGANIZATION AND LOCATION	DEGREE (if applicable)	RECEIPT DATE	FIELD OF STUDY
Argonne National Laboratory, Lemont, Illinois, United States	Postdoctoral Fellow	10/2009 - 12/2010	Nuclear Engineering
Tokyo Institute of Technology, Tokyo, Not Applicable, N/A, Japan	Postdoctoral Fellow	10/2008 - 10/2009	Nuclear Engineering
Tokyo Institute of Technology, Tokyo, Not Applicable, N/A, Japan	PHD	10/2005 - 10/2008	Nuclear Engineering
Politecnico di Milano, Milan, Not Applicable, N/A, Italy	MS	09/2003 - 07/2005	Nuclear Engineering
Politecnico di Milano, Milan, Not Applicable, N/A, Italy	BS	09/2000 - 07/2003	Energy Engineering

Appointments and Positions

2019 - present Professor, Pennyslvania State University, State College, Pennsylvania, United States

2010 - 2019 Principal Nuclear Engineer, Argonne National Laboratory, Lemont, Illinois, United States

Products

1. Merzari E, Hamilton S, Evans T, Min M, Fischer P, Kerkemeier S, Fang J, Romano P, Lan Y, Phillips M, Biondo E, Royston K, Warburton T, Chalmers N, Rathnayake T. Exascale Multiphysics Nuclear Reactor Simulations for Advanced Designs. Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis. SC '23: International Conference for High Performance Computing, Networking, Storage and Analysis; 12 1 23; Denver CO USA. New York, NY, USA: ACM; c2023. Available from: https://dl.acm.org/doi/10.1145/3581784.3627038 DOI: 10.1145/3581784.3627038
2. Fischer P, Kerkemeier S, Min M, Lan Y, Phillips M, Rathnayake T, Merzari E, Tomboulides A, Karakus A, Chalmers N, Warburton T. NekRS, a GPU-accelerated spectral element Navier–Stokes solver. Parallel Computing. 2022 December; 114:102982-. Available from: https://linkinghub.elsevier.com/retrieve/pii/S0167819122000710 DOI: 10.1016/j.parco.2022.102982
3. Offermans N, Peplinski A, Marin O, Merzari E, Schlatter P. Performance of Preconditioners for Large-Scale Simulations Using Nek5000. Lecture Notes in Computational Science and

Engineering [Internet] Cham: Springer International Publishing; 2020. Chapter Chapter 20263-272p. Available from: https://link.springer.com/10.1007/978-3-030-39647-3_20 DOI: 10.1007/978-3-030-39647-3_20

4. Hufnagel L, Canton J, Örlü R, Marin O, Merzari E, Schlatter P. The three-dimensional structure of swirl-switching in bent pipe flow. Journal of Fluid Mechanics. 2017 November 27; 835:86-101. Available from: https://www.cambridge.org/core/product/identifier/S0022112017007492/type/journal_article DOI: 10.1017/jfm.2017.749
5. Makarashvili V, Merzari E, Obabko A, Siegel A, Fischer P. A performance analysis of ensemble averaging for high fidelity turbulence simulations at the strong scaling limit. Computer Physics Communications. 2017 October; 219:236-245. Available from: https://linkinghub.elsevier.com/retrieve/pii/S0010465517301650 DOI: 10.1016/j.cpc.2017.05.023

Certification:

I understand that I have been designated as a covered individual by the Federal funding agency.

I certify to the best of my knowledge and belief that the information contained in this Biographical Sketch Form is true, complete, and accurate. I understand that any false, fictitious, or fraudulent information, misrepresentations, half-truths, or omissions of any material fact, may subject me to criminal, civil or administrative penalties for fraud, false statements, false claims or otherwise. (18 U.S.C. §§ 1001 and 287, and 31 U.S.C. 3729-3733 and 3801-3812). I further understand and agree that (1) the statements and representations made herein are material to DOE's funding decision, and (2) I have a responsibility to update the disclosures during the project period of the award should circumstances change which impact the responses provided above.

I also certify that, at the time of submission, I am not a party in a malign foreign talent recruitment program. I further understand should I take action to involve myself with a Malign Foreign Talent Recruitment Program during the period of performance of the award, I must notify the recipient's Authorized Agent immediately, but no later than five business days of taking such action and immediately recuse myself from all DOE awards.

Certified by Merzari, Elia in SciENcv on 2026-04-20 12:33:00

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors. Follow this format for each person.

*NAME: Merzari, Elia

PERSISTENT IDENTIFIER (PID): https://orcid.org/0000-0003-0174-7424

*POSITION TITLE: Professor

*ORGANIZATION AND LOCATION: Penn State, State College, Pennsylvania, United States

Proposals and Active Projects

***Proposal/Active Project Title:** Support for advanced modeling needs for Sawtooth

***Status of Support:** Current

Proposal/Award Number:

***Source of Support:** Sawtooth Simulations

***Primary Place of Performance:** Pennsylvania State University

***Proposal/Active Project Start Date: (MM/YYYY):** 01/2026

***Proposal/Active Project End Date: (MM/YYYY):** 12/2026

***Total Anticipated Proposal/Project Amount:** $171,013

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.1

***Overall Objectives:**

Support the modeling needs of Sawtooth simulation, focused on LWR (light water reactor) fuel assembly modeling.

***Statement of Potential Overlap:**

None.

***Proposal/Active Project Title:** Merzari ANL Joint Appointment 2025-2026

***Status of Support:** Current

Proposal/Award Number:

***Source of Support:** Argonne National Laboratory

***Primary Place of Performance:** Argonne National Laboratory

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2025

***Proposal/Active Project End Date: (MM/YYYY):** 08/2026

***Total Anticipated Proposal/Project Amount:** $35,653

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1

***Overall Objectives:**

Support Argonne National Laboratory and the NEAMS project through a joint appointment.

***Statement of Potential Overlap:**

None.

***Proposal/Active Project Title:** Validation Test of Radiation and Natural/Convection Thermal Models for eVinci NTR Integrated CFD Analysis

***Status of Support:** Current

Proposal/Award Number:

***Source of Support:** Westinghouse

***Primary Place of Performance:** Pennsylvania State University

***Proposal/Active Project Start Date: (MM/YYYY):** 11/2024

***Proposal/Active Project End Date: (MM/YYYY):** 10/2027

***Total Anticipated Proposal/Project Amount:** $376,269

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	0.1
2026	0.1
2026	0.1

***Overall Objectives:**

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Develop and experimental facility to validate the modeling of passive heat removal in the NTR.

***Statement of Potential Overlap:**

None.

***Proposal/Active Project Title:**	Center for Thermal-Fluids Applications in Nuclear Energy: Toward Industry Adoption
***Status of Support:**	Current
Proposal/Award Number:	DE-NE0009458
***Source of Support:**	DOE-NE (NEUP)
***Primary Place of Performance:**	Pennsylvania State University
***Proposal/Active Project Start Date: (MM/YYYY):**	08/2024
***Proposal/Active Project End Date: (MM/YYYY):**	09/2027
***Total Anticipated Proposal/Project Amount:**	$1,458,693

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	0.25
2026	0.25
2027	0.1

***Overall Objectives:**

Continue developing multiscale framework to model thermal-hydraulic phenomena in advanced reactors. Phase II focuses on transients.

***Statement of Potential Overlap:**

None

***Proposal/Active Project Title:**	CVAP Support

***Status of Support:** Current

Proposal/Award Number: 23-005.PSU

***Source of Support:** Structural Integrity Associates, Inc.

***Primary Place of Performance:** Pennsylvania State University

***Proposal/Active Project Start Date: (MM/YYYY):** 11/2023

***Proposal/Active Project End Date: (MM/YYYY):** 12/2026

***Total Anticipated Proposal/Project Amount:** $73,200

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.21

***Overall Objectives:**

Support the Fluid-Structure Interaction analysis of Natrium

***Statement of Potential Overlap:**

None

***Proposal/Active Project Title:** Reduced Order Modeling of Heat and Fluid Flow: Multi-Scale Modeling of Advanced Reactors to Enable Faster Deployment

***Status of Support:** Current

Proposal/Award Number: 115213-19749

***Source of Support:** University Of Illinois at Urbana-Champaign (Prime: DOE-NE)

***Primary Place of Performance:** Pennsylvania State University

***Proposal/Active Project Start Date: (MM/YYYY):** 10/2023

***Proposal/Active Project End Date: (MM/YYYY):** 09/2026

***Total Anticipated Proposal/Project Amount:** $300,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2024	0.23
2025	0.23

Year	Person Months
2026	0.23

***Overall Objectives:**

Develop a novel reduced order modeling approach for the modeling of fast reactors.

***Statement of Potential Overlap:**

None.

***Proposal/Active Project Title:**	An Innovative Monitoring Technology for the Reactor Vessel of Micro-HTGR
***Status of Support:**	Current
Proposal/Award Number:	M2300738
***Source of Support:**	Texas A&M University (Prime Sponsor: DOE-NE, NEUP)
***Primary Place of Performance:**	Pennsylvania State University
***Proposal/Active Project Start Date: (MM/YYYY):**	10/2022
***Proposal/Active Project End Date: (MM/YYYY):**	09/2026
***Total Anticipated Proposal/Project Amount:**	$225,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2023	0.21
2024	0.21
2025	0.21

***Overall Objectives:**

Developing a novel Machine-Learning based based sparse sensor methodology for advanced reactor modeling.

***Statement of Potential Overlap:**

None.

***Proposal/Active Project Title:**	NSF SAFE-OSE: Securing the Cardinal Open-Source Ecosystem for Trustworthy Computations of Advanced Energy Systems
***Status of Support:**	Pending
Proposal/Award Number:	
***Source of Support:**	NSF
***Primary Place of Performance:**	Pennsylvania State University
***Proposal/Active Project Start Date: (MM/YYYY):**	01/2027
***Proposal/Active Project End Date: (MM/YYYY):**	12/2028
***Total Anticipated Proposal/Project Amount:**	$1,500,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	1
2028	1

***Overall Objectives:**

The project aims to transform the Cardinal multiphysics simulation ecosystem into a secure, reproducible, and tamper-resistant reference platform for high-consequence scientific computing by embedding threat-driven protections across software builds, dependencies, runtime-generated kernels, artifact distribution, and governance.

***Statement of Potential Overlap:**

None.

***Proposal/Active Project Title:**	Small Modular and Advanced Reactor Training (SMART) Program
***Status of Support:**	Pending
Proposal/Award Number:	
***Source of Support:**	Pittsburgh Technical
***Primary Place of Performance:**	Pennsylvania State University
***Proposal/Active Project Start Date: (MM/YYYY):**	09/2026
***Proposal/Active Project End Date: (MM/YYYY):**	08/2028
***Total Anticipated Proposal/Project Amount:**	$1,000,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.55
2027	1.2
2028	1.2

***Overall Objectives:**

Develop micro-credentials to support workforce development in nuclear in the state of Pennsylvania.

***Statement of Potential Overlap:**

None.

***Proposal/Active Project Title:**	AI-Enabled Integrated Design and Licensing Workflow for the HORIZON Advanced Nuclear Facility at Pennsylvania State University
***Status of Support:**	Pending
Proposal/Award Number:	
***Source of Support:**	Department of Energy
***Primary Place of Performance:**	Pennsylvania State University
***Proposal/Active Project Start Date: (MM/YYYY):**	07/2026
***Proposal/Active Project End Date: (MM/YYYY):**	03/2027
***Total Anticipated Proposal/Project Amount:**	$750,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.48

***Overall Objectives:**

The project seeks to demonstrate an AI-driven design process for microreactors and demonstrate it on the HORIZON facility at Penn State.

***Statement of Potential Overlap:**

Little to no overlap. There may be synergies in terms of methods that could be established if both projects are funded.

***Proposal/Active Project Title:**	Foundation Models for Automated Design of Energy Flow Systems
***Status of Support:**	Pending
Proposal/Award Number:	
***Source of Support:**	Argonne National Laboratory
***Primary Place of Performance:**	Pennsylvania State University
***Proposal/Active Project Start Date: (MM/YYYY):**	07/2026
***Proposal/Active Project End Date: (MM/YYYY):**	03/2027
***Total Anticipated Proposal/Project Amount:**	$70,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.15

***Overall Objectives:**

Develop foundation models to support AI-driven optimization of energy system performance.

***Statement of Potential Overlap:**

None. There may synergies in terms of methods if both projects are funded.

***Proposal/Active Project Title:**	Optimization of Liquid Metal Blanket Designs in Fusion Reactors Utilizing High-fidelity Thermalhydraulic Modeling and Artificial Intelligence
***Status of Support:**	Pending
Proposal/Award Number:	
***Source of Support:**	Department of Energy
***Primary Place of Performance:**	Pennsylvania State University
***Proposal/Active Project Start Date: (MM/YYYY):**	07/2026
***Proposal/Active Project End Date: (MM/YYYY):**	03/2027
***Total Anticipated Proposal/Project Amount:**	$750,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.5

***Overall Objectives:**

The objective of this project is to develop an AI-enabled, multiphysics design and control framework for fusion breeding blankets and related components that combines high-fidelity simulations with graph-neural-network foundation models to rapidly predict, optimize, and improve tritium breeding, thermal performance, and durability while reducing design times from weeks to days.

***Statement of Potential Overlap:**

None.

***Proposal/Active Project Title:**	Simulation of Material Aging (SOMA) - Multiscale and multiphysics simulations for predictive time-to-failure models of structural fusion materials
***Status of Support:**	Pending
Proposal/Award Number:	
***Source of Support:**	Department of Energy
***Primary Place of Performance:**	Pennsylvania State University
***Proposal/Active Project Start Date: (MM/YYYY):**	07/2026
***Proposal/Active Project End Date: (MM/YYYY):**	03/2027
***Total Anticipated Proposal/Project Amount:**	$100,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.32

***Overall Objectives:**

Use curated experimental data and predictive material science simulation to develop an AI model that predicts time-to-failure for materials used in the structure of a fusion device, thereby accelerate the identification of viable structural materials.

***Statement of Potential Overlap:**

None.

***Proposal/Active Project Title:**	SPECTRA: A Sparse-sensing, Physics-Enhanced Control Twin for Reactor Autonomy

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** Department of Energy

***Primary Place of Performance:** Pennsylvania State University

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2026

***Proposal/Active Project End Date: (MM/YYYY):** 03/2027

***Total Anticipated Proposal/Project Amount:** $750,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.15

***Overall Objectives:**

We propose to develop the first holistic AI-enabled digital twin framework for microreactor autonomous operations integrating sparse fiber-optic sensing, cyber-awareness, and agentic AI control recommendations into a unified, continuously self-updating system.

***Statement of Potential Overlap:**

This proposal.

***Proposal/Active Project Title:** NSF POSE: Phase II: Cardinal: An Open-Source Ecosystem for High-Fidelity Multiphysics for Energy Applications

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** UIUC (Prime: NSF)

***Primary Place of Performance:** Pennsylvania State University

***Proposal/Active Project Start Date: (MM/YYYY):** 01/2026

***Proposal/Active Project End Date: (MM/YYYY):** 12/2027

***Total Anticipated Proposal/Project Amount:** $300,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.5

Year	Person Months
2027	0.5

***Overall Objectives:**

Develop a high-fidelity ecosystem for multi-physics simulation across multiple applications.

***Statement of Potential Overlap:**

None.

***Proposal/Active Project Title:**	Estimation of low temperature cladding failures during an RIA transient: Phase II Validation with HERA results
***Status of Support:**	Pending
Proposal/Award Number:	
***Source of Support:**	DOE-NE
***Primary Place of Performance:**	Pennsylvania State University
***Proposal/Active Project Start Date: (MM/YYYY):**	08/2025
***Proposal/Active Project End Date: (MM/YYYY):**	07/2027
***Total Anticipated Proposal/Project Amount:**	$533,333

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.25
2027	0.25

***Overall Objectives:**

Extend the work of Phase-I on the prediction of low-temperature RIA failures. The focus of this proposal is on validation with the HERA experiments.

***Statement of Potential Overlap:**

None.

***Proposal/Active Project Title:** Unraveling Laminar-Turbulent Transition in Rod Bundles

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** NEUP (Department of Energy)

***Primary Place of Performance:** Pennsylvania State University

***Proposal/Active Project Start Date: (MM/YYYY):** 08/2025

***Proposal/Active Project End Date: (MM/YYYY):** 07/2028

***Total Anticipated Proposal/Project Amount:** $600,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.12
2027	0.12
2028	0.12

***Overall Objectives:**

This project seeks to develop an experimental and numerical program to build a comprehensive theory of laminar-turbulent transition in rod bundles.

***Statement of Potential Overlap:**

None

***Proposal/Active Project Title:** Development and Commercialization of Multi-phase Modeling Modules for Accelerated Adoption of NEAMS in Nuclear Reactor Industry

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** Dynaflow

***Primary Place of Performance:** Pennsylvania State University

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2025

***Proposal/Active Project End Date: (MM/YYYY):** 03/2026

***Total Anticipated Proposal/Project Amount:** $60,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	0.55

***Overall Objectives:**

Develop a multiscale framework for the modeling of two-phase flow supporting advanced reactor applications.

***Statement of Potential Overlap:**

None

***Proposal/Active Project Title:**	The Nuclear Industry Readiness Certification Program: Safety Excellence for the Next-Generation Nuclear Workforce
***Status of Support:**	Pending
Proposal/Award Number:	
***Source of Support:**	DOE-NE
***Primary Place of Performance:**	Pennsylvania State University
***Proposal/Active Project Start Date: (MM/YYYY):**	06/2025
***Proposal/Active Project End Date: (MM/YYYY):**	05/2027
***Total Anticipated Proposal/Project Amount:**	$4,000,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	1
2026	1

***Overall Objectives:**

Develop a certification and related certificate program focused on nuclear professionalism.

***Statement of Potential Overlap:**

None.

***Proposal/Active Project Title:**	Identifying Risk and Reducing Uncertainty in Accident Scenarios During the Approach to Equilibrium for Pebble Bed Reactors
***Status of Support:**	Pending
Proposal/Award Number:	
***Source of Support:**	Idaho National Laboratory
***Primary Place of Performance:**	Pennsylvania State University
***Proposal/Active Project Start Date: (MM/YYYY):**	04/2025
***Proposal/Active Project End Date: (MM/YYYY):**	03/2027
***Total Anticipated Proposal/Project Amount:**	$209,855

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	0.1
2026	0.1

***Overall Objectives:**

Understand Uncertainty in Accident Scenarios during the approach to equilibrium for Pebble Bed Reactors through modeling and simulation.

***Statement of Potential Overlap:**

None.

Addendum

Do the Notice of Funding Opportunity (NOFO) or Award instructions require a Current and Pending (Other) Support addendum?

No

Certification:

I understand that I have been designated as a covered individual by the Federal funding agency.

I certify to the best of my knowledge and belief that the information contained in this Current and Pending (Other) Support Form is true, complete, and accurate. I understand that any false, fictitious, or fraudulent information, misrepresentations, half-truths, or omissions of any material fact, may subject me to criminal, civil or administrative penalties for fraud, false statements, false claims or otherwise. (18 U.S.C. §§ 1001 and 287, and 31 U.S.C. 3729-3733 and 3801-3812). I further understand and agree that (1) the statements and representations made herein are material to DOE's funding decision, and (2) I have a responsibility to update the disclosures during the project period of the award should circumstances change which impact the responses provided above.

I also certify that, at the time of submission, I am not a party in a malign foreign talent recruitment program. I further understand should I take action to involve myself with a Malign Foreign Talent Recruitment Program during the period of performance of the award, I must notify the recipient's Authorized Agent immediately, but no later than five business days of taking such action and immediately recuse myself from all DOE awards.

If this is an R&D proposal/award, I also certify to the following statement:

Within the past 12 months I have completed research security training meeting the requirements in SEC. 10634(b) of 42 USC 19234.

Certified by Merzari, Elia in SciENcv on 2026-04-20 14:24:48

SYNERGISTIC ACTIVITIES

NAME: Elia Merzari

ORCID ID: 0000-0003-0174-742

POSITION TITLE: Professor, Department of Nuclear Engineering

PRIMARY ORGANIZATION AND LOCATION: The Pennsylvania State University, University Park, PA

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1. American Nuclear Society Fellow, 2023
2. American Society of Mechanical Engineers Fellow, 2021
3. Untermeyer & Cisler Reactor Technology Medal (American Nuclear Society), 2025
4. George Westinghouse Silver Medal (ASME), 2016
5. R&D100 award (2016, 2023)

RESEARCH & RELATED BUDGET - Budget Period 1

OMB Number: 4040-0001
Expiration Date: 01/31/2029

UEI: NPM2J7MSCF61 **Enter name of Organization:** The Pennsylvania State University

Budget Type: ☐ Project ☒ Subaward/Consortium **Budget Period: 1** **Start Date:** 07/01/2026 **End Date:** 03/31/2027

A. Senior/Key Person

Prefix	First	Middle	Last	Suffix	Base Salary ($)	Months Cal.	Months Acad.	Months Sum.	Requested Salary ($)	Fringe Benefits ($)	Funds Requested ($)
Dr.	Elia		Merzari					0.15	3,047.00	1,237.00	4,284.00

Project Role: PD/PI

Additional Senior Key Persons: Add Attachment Delete Attachment View Attachment

Total Funds requested for all Senior Key Persons in the attached file

Total Senior/Key Person 4,284.00

B. Other Personnel

Number of Personnel	Project Role	Months Cal.	Months Acad.	Months Sum.	Requested Salary ($)	Fringe Benefits ($)	Funds Requested ($)
	Post Doctoral Associates						
1	Graduate Students		3.75		24,750.00	2,772.00	27,522.00
	Undergraduate Students						
	Secretarial/Clerical						

1 **Total Number Other Personnel**

Total Other Personnel 27,522.00

Total Salary, Wages and Fringe Benefits (A+B) 31,806.00

C. Equipment Description

List items and dollar amount for each item exceeding $5,000

Equipment item	Funds Requested ($)

Additional Equipment: Add Attachment | Delete Attachment | View Attachment

Total funds requested for all equipment listed in the attached file	
Total Equipment	

D. Travel

	Funds Requested ($)
1. Domestic Travel Costs (Incl. Canada, Mexico and U.S. Possessions)	10,000.00
2. Foreign Travel Costs	
Total Travel Cost	10,000.00

E. Participant/Trainee Support Costs

	Funds Requested ($)
1. Tuition/Fees/Health Insurance	
2. Stipends	
3. Travel	
4. Subsistence	
5. Other	
Number of Participants/Trainees — **Total Participant/Trainee Support Costs**	

F. Other Direct Costs

		Funds Requested ($)
1.	Materials and Supplies	
2.	Publication Costs	
3.	Consultant Services	
4.	ADP/Computer Services	
5.	Subawards/Consortium/Contractual Costs	
6.	Equipment or Facility Rental/User Fees	
7.	Alterations and Renovations	
8.	Tuition	4,242.00
9.		
10.		
11.		
12.		
13.		
14.		
15.		
16.		
17.		
	Total Other Direct Costs	4,242.00

G. Direct Costs

	Funds Requested ($)
Total Direct Costs (A thru F)	46,048.00

H. Indirect Costs

Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base ($)	Funds Requested ($)
MTDC	57.30	41,806.00	23,952.00

Total Indirect Costs 23,952.00

Cognizant Federal Agency
(Agency Name, POC Name, and POC Phone Number)

ONR,Boston Office,E-mail: usn.pentagon.cnr-arlington-va.mbx.onr-boston@us.navy.mil

I. Total Direct and Indirect Costs

	Funds Requested ($)
Total Direct and Indirect Institutional Costs (G + H)	70,000.00

J. Fee

	Funds Requested ($)

K. Total Costs and Fee

	Funds Requested ($)
Total Costs and Fee (I + J)	70,000.00

L. Budget Justification

(Only attach one file.) EM Budget Justification Final.pdf

Add Attachment | Delete Attachment | View Attachment

RESEARCH & RELATED BUDGET - Cumulative Budget

		Totals ($)
Section A, Senior/Key Person		4,284.00
Section B, Other Personnel		27,522.00
Total Number Other Personnel	1	
Total Salary, Wages and Fringe Benefits (A+B)		31,806.00
Section C, Equipment		
Section D, Travel		10,000.00
1. Domestic	10,000.00	
2. Foreign		
Section E, Participant/Trainee Support Costs		
1. Tuition/Fees/Health Insurance		
2. Stipends		
3. Travel		
4. Subsistence		
5. Other		
6. Number of Participants/Trainees		
Section F, Other Direct Costs		4,242.00
1. Materials and Supplies		
2. Publication Costs		
3. Consultant Services		
4. ADP/Computer Services		
5. Subawards/Consortium/Contractual Costs		
6. Equipment or Facility Rental/User Fees		
7. Alterations and Renovations		
8. Other 1	4,242.00	
9. Other 2		
10. Other 3		
11. Other 4		
12. Other 5		
13. Other 6		
14. Other 7		
15. Other 8		
16. Other 9		
17. Other 10		

Section G, Direct Costs (A thru F)	46,048.00
Section H, Indirect Costs	23,952.00
Section I, Total Direct and Indirect Costs (G + H)	70,000.00
Section J, Fee	
Section K, Total Costs and Fee (I + J)	70,000.00

BUDGET JUSTIFICATION
The Pennsylvania State University

(A-B.) Personnel ($27,797) – The principal investigator is budgeted at the percentage of time shown using his/her actual salary in the calculation. The principal investigator's time includes both technical and project management functions. Any other individuals/positions shown are technical staff with the percentage of time shown and actual salaries used. For project time occurring after July 1 of any given year, the salaries have been adjusted at the University approved rate of 4%.

Senior/Key Personnel ($3,047):

- **Elia Merzari, Principal Investigator** – 9.9% effort over 1.5 months (.15 summer months) for the project. Elia Merzari will support the project as the PI at Penn State, interacting with project leadership and supervising the activities of the graduate student. He will also help identify and define benchmark problems relevant to nuclear energy and support the broader goals of the project.

Other Personnel ($24,750):

- **To Be Named, Graduate Assistant** – One Grade 16 Graduate Assistant @ 50% effort over 7.5 months (3.75 academic months) for the project. The student will perform the majority of the work at Penn State, focusing on the development of high-fidelity numerical models relevant to nuclear energy and on testing the novel foundation-model approaches developed in the project. The student will also run simulations, post-process results, and prepare documents, presentations, and reports as required by project leadership.
  ** Graduate Assistants at Penn State University are considered full-time at 50% effort with the expectation that the remaining 50% is dedicated to coursework.*

Fringe Benefits ($4,009) – Fringe benefits are computed using the fixed rates of 40.6% applicable to Category I Salaries, 11.2% applicable to Category II Graduate Assistants, 7.9% applicable to Category III Wages, 0.3% applicable to Category IV Student Wages, and 27.7% for Category V, Postdoctoral Scholars and Fellows, for fiscal year 2026 (July 1, 2025, through June 30, 2026). If this proposal is funded, the rates quoted above shall, at the time of funding, be subject to adjustment for any period subsequent to June 30, 2026, if superseding Government approved rates have been established. Fringe benefit rates are negotiated and approved by the Office of Naval Research, Penn State's cognizant federal agency.

(C.) Equipment ($0)

(D.) Travel ($10,000) – All travel will be in accordance with University travel regulations and mileage will be charged at the current rate on the date of travel. Travel estimates are based on costs that were incurred on previous projects of a similar nature for federal and state agencies.

1. **Domestic Travel ($10,000)** – Funding is requested for the PI to attend the required two annual Genesis Mission PI Meetings. The PI will participate in coordination activities with other projects and to effectively communicate with other researchers. Location TBD.

Year	Destination	Dates	# Travelers	Cost per Traveler	Total Estimated Costs	Purpose/Relation to Project	Basis for Cost Estimate
2026	TBD	TBD	1	$5,000	$5,000	annual PI meeting	Prior travel
2027	TBD	TBD	1	$5,000	$5,000	annual PI meeting	Prior travel

(E.) Participant/Trainee Support Costs ($0)

(F.) Other Direct Costs

1. **Materials and Supplies ($0)**
2. **Publication Costs ($0)**
3. **Consultant Services ($0)**
4. **ADP/Computer Services ($0)**
5. **Subawards/Consortium/Contractual Costs ($0)**
6. **Equipment or Facility Rental/User Fees ($0)**
7. **Alterations and Renovations ($0)**
8. **Other ($4,242)**:
    - **Graduate Assistant Tuition ($4,242)** – Computed using the approved tuition charges for a one-half (1/2) time graduate assistant of $10,810 for fall semester 2025, $10,810 for spring semester 2026, and $5,515 for summer session 2026. The charges quoted above are increased by two percent (2%) for any project period occurring after summer session 2026, and each summer session thereafter. NOTE*:* It is anticipated that the graduate student assigned to this project will be classified as a post-comprehensive [level 601] PhD student, therefore a reduced dissertation fee has been budgeted for this project.
      ** Tuition is included for Graduate Assistants for Fall and Spring Semesters.*

(G.) Total Direct Costs (A-F)
Year 1: $46,048; Total: $46,048

(H.) Indirect Costs (Facilities and Administrative Costs) ($23,952) – F&A rates are negotiated and approved by the Office of Naval Research, Penn State's cognizant federal agency. Penn State's current fixed on-campus rate for research is 57.3% of MTDC from July 1, 2025, through June 30, 2026. New awards and new competitive segments with an effective date of July 1, 2026, or later shall be subject to adjustment when superseding Government approved rates are established. Per 2 CFR 200 (Appendix III, Section C.7), the actual F&A rates used will be fixed at the time of the initial award for the duration of the competitive segment.

Cognizant Agency: Office of Naval Research
Contact Information: Boston Regional Office
495 Summer Street
Boston, MA 02210
Email: usn.pentagon.cnr-arlington-va.mbx.onr-boston@us.navy.mil

Year 1: $23,952; Total: $23,952

(I.) Total Direct and Indirect Costs (G-H)
Year 1: $70,000; Total: $70,000

The Pennsylvania State University employs individuals and accepts students and graduate research students from a multitude of national backgrounds. As an entity, the University is subject to, and works diligently to obey, federal regulations regarding the export of controlled technologies and data. Sponsor, as an independent entity, is individually responsible for ascertaining its compliance with federal export laws and procedures. If Sponsor anticipates disclosure or provision of controlled technology or data to University as part of the proposed sponsored project, Sponsor should inform University, in writing, of the existence of, and information concerning the scope and extent of, such anticipated disclosures or provisions.

DEPARTMENT OF THE NAVY
OFFICE OF NAVAL RESEARCH
875 NORTH RANDOLPH STREET
SUITE 1425
ARLINGTON, VA 22203-1995

IN REPLY REFER TO:

February 26, 2026
[Supersedes Rate Agreement dated June 10, 2025]

NEGOTIATION AGREEMENT

INSTITUTION: **THE PENNSYLVANIA STATE UNIVERSITY**
UNIVERSITY PARK, PA 16801-3857

The Facilities and Administrative (F&A) cost rates contained herein are for use on grants, contracts and/or other agreements issued or awarded to the Pennsylvania State University (PSU) by all Federal Agencies of the United States of America, in accordance with the provisions and cost principles mandated by 2 CFR Part 200. These rates shall be used for forward pricing and billing purposes for the PSU's Fiscal Year 2026. This rate agreement supersedes all previous rate agreements/determinations related to these rates for Fiscal Year 2026.

SECTION I – RATES – TYPE: FIXED (FIXED)

UNIVERSITY PARK (F&A)

TYPE	FROM	TO	RATE	BASE	APPLICABLE TO	LOCATION
FIXED	07/01/25	06/30/26	57.3%	(a)	Organized Research (1,4)	On-Campus
FIXED	07/01/25	06/30/26	26.0%	(a)	Organized Research (1,4)	Off-Campus
FIXED	07/01/25	06/30/26	64.4%	(a)	Organized Research (2,4)	On-Campus
FIXED	07/01/25	06/30/26	33.1%	(a)	Organized Research (2,4)	Off-Campus
FIXED	07/01/25	06/30/26	53.3%	(a)	Instruction (1,4)	On-Campus
FIXED	07/01/25	06/30/26	26.0%	(a)	Instruction (1,4)	Off-Campus
FIXED	07/01/25	06/30/26	24.1%	(a)	OSA[1] (1,4)	On-Campus
FIXED	07/01/25	06/30/26	16.8%	(a)	OSA (1,4)	Off-Campus
FIXED	07/01/25	06/30/26	23.7%	(a)	OSA HHS (1,3)	On-Campus
FIXED	07/01/25	06/30/26	16.4%	(a)	OSA HHS (1,3)	Off-Campus

APPLIED RESEARCH LABORATORY

TYPE	FROM	TO	RATE	BASE	APPLICABLE TO	LOCATION
FIXED	07/01/25	06/30/26	6.5%	(b)	Organized Research – ARL[2]	On-Campus
FIXED	07/01/25	06/30/26	5.3%	(b)	Organized Research - ARL	Off-Campus

HERSHEY COLLEGE OF MEDICINE

TYPE	FROM	TO	RATE	BASE	APPLICABALE TO	LOCATION
FIXED	07/01/25	06/30/26	56.6%	(a)	Organized Research – HCM[3]	On-Campus
FIXED	07/01/25	06/30/26	26.0%	(a)	Organized Research - HCM	Off-Campus

[1]Other Sponsored Activities
[2]Applied Research Laboratory
[3] Hershey College of Medicine

DISTRIBUTION BASE

(a) Modified Total Direct Cost (MTDC) means all direct salaries and wages, applicable fringe benefits, materials and supplies, services, travel, and up to the first $50,000 of each subaward (regardless of the period of performance of the subawards under the award). MTDC excludes equipment, capital expenditures, charges for patient care, rental costs, tuition remission, scholarships and fellowships, participant support costs, and the portion of each subaward in excess of $50,000.

(b) The ARL sponsored base consists of all on-site and off-site direct costs recorded to projects, (excluding direct material and equipment related costs and direct subcontract costs and tuition). The ARL internal overhead base consists of all on-site and off-site ARL internal overhead costs and F&A is allocated on all direct costs (excludes depreciation, rent, tuition, and allowable interest).

APPLICABLE TO

(1) Applies to DOD contracts awarded before November 30, 1993, all Non-DOD Instruments, and all DOD grants and other agreements (See Section II, paragraph E). (Capped).

(2) Applies to only DOD contracts awarded on or after November 30, 1993 in accordance with and under the authority of DFARS 231.303(1) (See Section II, paragraph E). (Uncapped).

(3) Applies to all awards subject to Executive Level II Salary Limitations as defined in the provisions of HHS Appropriation Act (PUB.L. No. 117-328, div. H, title II) or similar provision in subsequent appropriations acts.

(4) Applies to all awards not subject to the HHS Appropriation.

SECTION II - GENERAL TERMS AND CONDITIONS

A. **LIMITATIONS**: Use of the rates set forth under Section I is subject to availability of funds and to any other statutory or administrative limitations. The rates are applicable to a given grant, contract or other agreement only to the extent that funds are available and consistent with any and all limitations of cost clauses or provisions, if any, contained therein. Acceptance of any or all of the rates agreed to herein is predicated upon the following conditions: (1) that no

costs other than those incurred by the institution were included in this indirect cost pool as finally accepted and that such costs are legal obligations of the institution and allowable under governing cost principles; (2) that the same costs that have been treated as indirect costs are not claimed as direct costs; (3) that similar types of costs have been accorded consistent accounting treatment; and (4) that the information provided by the institution which was used as a basis for acceptance of the rates agreed to herein, and expressly relied upon by the Government in negotiating and accepting the said rates is not subsequently found to be materially incomplete or inaccurate.

B. **ACCOUNTING CHANGES**: The rates contained in Section I of this agreement are based on the accounting system in effect at the time the agreement was negotiated. Changes to the method(s) of accounting for costs, which affect the amount of reimbursement resulting from the use of these rates require the prior written approval of the authorized representative of the cognizant agency for indirect costs. Such changes include but are not limited to changes in the charging of a particular type of cost from indirect to direct. Failure to obtain such approval may result in subsequent cost disallowances.

C. **FIXED RATES WITH CARRY-FORWARD PROVISIONS**: The fixed rates contained in this agreement are based on estimates of the costs for FY 2026. When actual costs for this fiscal year are determined, adjustments will be applied to a rate negotiation for a subsequent fiscal year to recognize the difference between the FY 2026 estimated costs used to establish the fixed rates and the negotiated actual FY 2026 costs.

D. **CARRY-FORWARD AMOUNTS**: The following estimated CFW for FY24 carry-forward amounts were included in the establishment of the FY2026 rates: ( ) Denotes over-recovery.

FY24 Estimated CFW

Organized Research – University Park	Liquidated in FY 26
Organized Research On-Campus Capped	($3,896,035)
Instruction On Campus	$17,773,801
OSA On Campus	($1,214,696)
OSA Off Campus	($28,851)
ARL On Campus	$205,096
ARL Off Campus	($15,152)
Hershey	($2,812,595)

E. **USE BY OTHER FEDERAL AGENCIES**: The rates set forth in Section I are negotiated in accordance with and under the authority set forth in 2 CFR Part 200. Accordingly, such rates shall be applied to the extent provided in such regulations to grants, contracts, and other agreements to which 2 CFR Part 200 applies, subject to any limitations in part A of this section. Copies of this document may be provided by either party to other federal agencies to provide such agencies with documentary notice of this agreement and its terms and conditions.

F. **DFARS WAIVER**: Signature of this agreement by the authorized representative of Pennsylvania State University and the Government acknowledges and affirms the University's request to waive the prohibition contained in DFARS 231.303(1) and the Government's exercise of its discretion contained in DFARS 231.303(2) to waive the prohibition in DFARS 231.303(1) with regards to all the rates except F&A Organized Research. The waiver request by Pennsylvania State University is made to simplify the University's overall management of DoD cost reimbursements under DoD contracts.

G. **SPECIAL REMARKS**:

1. The rates included in Section I are not intended to be applied to Intergovernmental Personnel Act (IPA) costs. If the Pennsylvania State University elects to seek reimbursement of F&A or internal overhead costs associated with IPA agreements, then the University and the Office of Naval Research shall establish special F&A and/or internal overhead rates for IPA agreements in accordance with the provisions of 2 CFR Part 200.
2. The Government's agreement with the rates set forth in Section I is not an acceptance of Pennsylvania State University's accounting practices or methodologies. Any reliance by the Government on cost data or methodologies submitted by Pennsylvania State University is on a non-precedence-setting basis and does not imply Government acceptance.

Accepted:

FOR PENNSYLVANIA STATE UNIVERSITY:	FOR THE U.S. GOVERNMENT:
Sara Thorndike Senior Vice President for Finance & Business/Treasurer	Betty J. Tingle Contracting Officer
2/26/2026 Date	Date

For information concerning this agreement contact:

Betty Tingle, Contracting Officer
Office of Naval Research

Phone: (948) 215-0521
E-mail: betty.j.tingle.civ@us.navy.mil

DEPARTMENT OF THE NAVY
OFFICE OF NAVAL RESEARCH
875 NORTH RANDOLPH STREET
SUITE 1425
ARLINGTON, VA 22203-1995

IN REPLY REFER TO:

Agreement Date: June 27, 2025
[Supersedes Agreement Dated: June 27, 2024]

NEGOTIATION AGREEMENT

INSTITUTION: **THE PENNSYLVANIA STATE UNIVERSITY**
UNIVERSITY PARK, PA 16801-3857

The Fringe Benefit rates contained herein are for use on grants, contracts and/or other agreements issued or awarded to the Pennsylvania State University (PSU) by all Federal agencies of the United States of America in accordance with the provisions and cost principles mandated by 2 CFR Part 200. These rates shall be used for forward pricing and billing purposes for the PSU's Fiscal Years 2026 and 2027. This rate agreement supersedes all previous rate agreements/determinations related to these rates for Fiscal Year 2026.

Section I: RATES - TYPE: FIXED (Fixed) and PROVISIONAL (Prov)

TYPE	FROM	TO	RATE	BASE	APPLICABLE TO	LOCATION
Fixed	7/1/2025	6/30/2026	40.6%	(a)	Salaries	All
Fixed	7/1/2025	6/30/2026	7.9%	(b)	Wages	All
Fixed	7/1/2025	6/30/2026	11.2%	(c)	Graduate Assistants	All
Fixed	7/1/2025	6/30/2026	0.3%	(d)	Students	All
Fixed	7/1/2025	6/30/2026	27.7%	(e)	Postdoctoral Students	All
Prov	7/1/2026	6/30/2027	40.6%	(a)	Salaries	All
Prov	7/1/2026	6/30/2027	7.9%	(b)	Wages	All
Prov	7/1/2026	6/30/2027	11.2%	(c)	Graduate Assistants	All
Prov	7/1/2026	6/30/2027	0.3%	(d)	Students	All
Prov	7/1/2026	6/30/2027	27.7%	(e)	Postdoctoral Students	All

DISTRIBUTION BASES

(a) Salaries, excluding graduate assistants, postdoctoral scholars, fixed term II and visiting faculty appointments not receiving full benefits.
(b) Wages and those salaries not receiving full benefits, exclusive of graduate assistants, student wages, and postdoctoral scholar salaries.
(c) Graduate assistant salaries.
(d) Student wages.
(e) Postdoctoral Scholar salaries.

SECTION II - GENERAL TERMS AND CONDITIONS

A. **LIMITATIONS:** Use of the rates set forth under Section I is subject to availability of funds and to any other statutory or administrative limitations (e.g., award term with indirect rate cap). The rates are applicable to a given grant, contract or other agreement only to the extent that funds are available and consistent with any and all limitations of cost clauses or provisions, if any, contained therein. Acceptance of any or all of the rates agreed to herein is predicated upon the following conditions: (1) that no costs other than those incurred by the institution were included in this indirect cost pool as finally accepted and that such costs are legal obligations of the institution and allowable under governing cost principles; (2) that the same costs that have been treated as indirect costs are not claimed as direct costs; (3) that similar types of costs have been accorded consistent accounting treatment; and (4) that the information provided by the institution which was used as a basis for acceptance of the rates agreed to herein, and expressly relied upon by the Government in negotiating and accepting the said rates is not subsequently found to be materially incomplete or inaccurate.

B. **ACCOUNTING CHANGES**: The rates contained in Section I of this agreement are based on the accounting system in effect at the time the agreement was negotiated. Changes to the method(s) of accounting for costs, which affect the amount of reimbursement resulting from the use of these rates require the prior written approval of the authorized representative of the cognizant agency for indirect costs. Such changes include but are not limited to changes in the charging of a particular type of cost from indirect to direct. Failure to obtain such approval may result in subsequent cost disallowances.

C. **FIXED RATES WITH CARRY-FORWARD PROVISIONS:** The fixed rates in this agreement are based on estimates of the costs for FY 2026. When actual costs for this fiscal year are determined, adjustments will be applied to a rate negotiation for a subsequent fiscal year to recognize the difference between the FY 2026 estimated costs used to establish the fixed rates and the negotiated actual FY 2026 costs.

D. **CARRY FORWARD AMOUNTS**: The estimated carry-forwards for FY24 were included in the establishment of the FY 2026 rates:

Liquidated in FY26

Employee Category	FY24 Estimated CFW Liquidated in FY26 Fixed Rates
Salaries	$23,330,634
Wages	$1,695
Graduate Assistants	$192,516
Student Wages	8,377
Postdoctoral Scholar Salaries	$50,565

E. **PROVISIONAL RATES:** The provisional rates contained in this agreement are subject to unilateral amendment by the Government or bilateral amendment by the contracting parties at any time.

F. **USE BY OTHER FEDERAL AGENCIES**: The rates set forth in Section I are negotiated in accordance with and under the authority set forth in 2 CFR Part 200. Accordingly, such rates shall be applied to the extent provided in such regulations to grants, contracts, and other agreements to which 2 CFR Part 200 applies, subject to any limitations in part A of this section. Copies of this document may be provided by either party to other federal agencies to provide such agencies with documentary notice of this agreement and its terms and conditions.

Accepted:

FOR PENNSYLVANIA STATE UNIVERSITY:	FOR THE U.S. GOVERNMENT:
Sara Thorndike, Senior Vice President for Finance & Business/Treasurer	Betty J. Tingle Contracting Officer
7/1/25 Date	Date

For information concerning this agreement contact:
Betty Tingle, Contracting Officer, Phone: (703) 696-7742, E-mail: betty.j.tingle.civ@us.navy.mil

OMB Number: 4040-0010
Expiration Date: 12/31/2026

Project/Performance Site Location(s)

Project/Performance Site Primary Location

☐ I am submitting an application as an individual, and not on behalf of a company, state, local or tribal government, academia, or other type of organization.

Organization Name: The Pennsylvania State University

UEI: NPM2J7MSCF61

* Street1: Office of Sponsored Programs

Street2: 200 Innovation Blvd., Suite 110

* City: University Park

County: Centre

* State: PA: Pennsylvania

Province:

* Country: USA: UNITED STATES

* ZIP / Postal Code: 16802-7000

* Project/ Performance Site Congressional District: PA-015

Project/Performance Site Location 1

☐ I am submitting an application as an individual, and not on behalf of a company, state, local or tribal government, academia, or other type of organization.

Organization Name:

UEI:

* Street1:

Street2:

* City:

County:

* State:

Province:

* Country: USA: UNITED STATES

* ZIP / Postal Code:

* Project/ Performance Site Congressional District:

Additional Location(s) Add Attachment Delete Attachment View Attachment

This format is provided for the convenience of the entity providing the disclosure and certification; however, the entity is not required to use this specific format. If another format is used, the signatory must include the same substantive information, a signature, date, and the certification statement provided in Section 3 below.

DEPARTMENT OF ENERGY
TRANSPARENCY OF FOREIGN CONNECTIONS DISCLOSURE AND CERTIFICATION

For the Authorized Official of the entity (required for both the recipient and subrecipients)

The following information is required as part of the Department of Energy's (DOE) research, technology and economic security due diligence review process. Each entity must disclose the following information as it relates to its respective entity. Include a separate disclosure and signed certification for each entity at the recipient and subrecipient level.

All items below require a response, unless exempt by entity type. If supplemental documentation is provided, identify the applicability to each item below. Supplemental information, including links to financial statements or annual reports, is not a replacement for responses. If you have nothing to report for a specific item, include "Nothing to Report" for that item.

Responses to disclosure questions may contain trade secrets or commercial or financial information that is privileged or confidential and is exempt from public disclosure. Such information shall be used or disclosed only for evaluation purposes or in accordance with an award between the submitter and the United States Government. See Appendix B for information on how to identify confidential information.

Save each TFC in a single file using the following convention for the title:
"BusinessSensitive_AwardNumber_EntityName(Prime or Sub)_TFC".

DOE reserves the right to request additional or clarifying information based on the information submitted.

Disclosure exceptions by entity type:

- DOE National Laboratories and domestic government entities are not required to respond to the Transparency of Foreign Connections disclosure.
- Institutions of higher education are only required to respond to items with an asterisk symbol (*).
- The applicability of the disclosure requirements is determined by the entity type. Regardless of whether the recipient is exempt, the subrecipient(s) must provide these disclosures unless the subrecipient is also exempt.

Questions: Contact rtesinfo@hq.doe.gov

*Section 1. Identifying Information

a. *Complete Legal Name of the Entity	The Pennsylvania State University
b. *Entity's Unique Entity ID (UEI)	NPM2J7MSCF61
c. *Entity's Website Address (link)	https://researchsupport.psu.edu/osp
d. *Entity's Mailing Address	200 Innovation Blvd., Suite 110 University Park, PA 16802-7000
e. *Entity's Physical Address	101 Technology Center Building University Park, PA 16802
f. *Entity's Business Point of Contact (POC) Name	Robin B. Riglin
g. *POC Phone Number	814-865-1372
h. *Project Title	Foundation Models for Automated Design of Energy Flow Systems
i. *Proposal # or Award #	260244

Section 2. Disclosure of Foreign Connections

1. ***Malign Foreign Talent Recruitment Program**
   *Is any principal investigator, project manager, or other covered individual, or business owner, officer party to a Malign Foreign Talent Recruitment Program (MFTRP)? ☐ Yes ☒ No

   *If yes, provide the following information: Full name, position, role in the project, name of the MFTRP, and dates of participation.

2. **Ownership Structure**
   a. Provide a complete capitalization table for your entity, including all equity interests (including LLC and partnership interests, as well as derivative securities). Include both the number of shares issued to each equity holder, as well as the percentage of that series and of all equity on a fully diluted basis. Identify the principal place of incorporation (or organization) for each equity holder. If the equity holder is a natural person, identify the citizenship(s).

- If the recipient or subrecipient is a publicly traded company, provide the above information for shareholders with an interest greater than five percent.

b. For any owner/investor identified that is an investment fund utilizing a general partner structure, identify: 1) the general partner or equivalent individual/entity; and 2) the place of incorporation and principal place of business if the general partner is an entity OR citizenship(s) if the general partner is a natural person.

c. Provide an entity chart (including ownership percentages and officers) to illustrate the relationship between your entity, your entity's immediate parents, intermediate parents, ultimate parents, and any subsidiaries.

   i. For your entity's ultimate parent, please identify all owners with greater than a five percent ownership interest.
   ii. For entities, please identify the locations of incorporation and principal places of business. For individuals, please identify countries of citizenship.

d. Are any entities, owners, or investors within your entity's ownership chain identified on the Bureau of Industry and Security Entity List (search via Consolidated Screening List and select "Entity List" for "Source"), Department of Defense 1260H List, and/or Department of the Treasury Office of Foreign Assets Control Treasury's Specially Designated Nationals list? ☐ Yes ☐ No

If yes, identify entity, owner or investor and the specific list.

e. Provide a summary table identifying all rounds of financing, the purchase dates, the investors for each round, and all of the associated governance and information rights obtained by investors during each round of financing.

3. Is your entity wholly or partially owned by an entity incorporated or otherwise formed in a foreign country of concern or foreign state-owned entity?
☐ Yes ☐ No

If yes, disclose the entity name, full address, and percentage of ownership.

4. Is there a parent company, joint venture, or subsidiary, of your entity that is based in or receives funding from any foreign country of concern? ☐ Yes ☐ No

   If yes, disclose the entity name, full address, applicant or awardee relationships (i.e., parent company, joint venture, subsidiary, etc.) of each entity based in, or funded by, any foreign country of concern.

5. Does your entity have any current or pending contractual or financial obligation or other agreement specific to a business arrangement, or joint venture-like arrangement with an entity based in, or funded by, any foreign country of concern? ☐ Yes ☐ No

   If yes, proceed to question 5a. If no, proceed to question 6.

   5a. Is the obligation, agreement, or arrangement connected to the project identified in Section 1?
   ☐ Yes ☐ No

   If yes, disclose the name of each enterprise or foreign entity, type of obligation, agreement, or arrangement (i.e., contractual, financial, or other), description of obligation, agreement, or arrangement, and the foreign state(s) and/or the country of the foreign entity (or entities).

6. Does your entity have any venture capital or institutional investment? ☐ Yes ☐ No

   If yes, proceed to question 6a. If no, proceed to question 7.

   6a. Does the investing entity have a general partner or any other individual holding a leadership role who has a foreign affiliation with a foreign country of concern?
   ☐ Yes ☐ No ☐ Unable to determine

If yes or unable to determine, disclose the venture capital or institutional investing entity's name, the percentage of ownership obtained by the investing entity, and the type of investment (i.e., equity, debt, or combination of equity and debt).

7. *During the previous 5-year period, did your entity have any technology licensing or intellectual property sales or transfers to a foreign country of concern? ☒ Yes ☐ No

   If yes, proceed to question 7b. If no, proceed to question 8.
   *7b. Was the underlying technology in the same technology area (e.g., batteries, biotechnology, grid, energy generation and storage, advanced computing) as the project identified in Section 1? ☐ Yes ☒ No

   *If yes, disclose the name, address, and country, of the institution or entity that licensed, purchased, or received the technology or intellectual property.

Licensee or Intellectual Property Buyer Name	Foreign Country of Concern	Full Street Address
Shanghai Bovax Biotechnology Co., Ltd.	**China**	**Shanghai Bovax Biotechnology Co., Ltd. Building 12, No. 908, Ziping Road Pudong New District Shanghai, 201231, China**

8. Is there any foreign business entity, offshore entity, or entity outside the United States related to your entity? ☐ Yes ☐ No

   If yes, disclose the entity name, relationship type (i.e., foreign business entity, offshore entity, entity outside the United States), description of the relationship to the applicant or awardee, and entity address and country.

9. Provide a complete list of all directors (and board observers), including their full name, shareholder affiliation, date of appointment, voting rights, citizenship, duration of term, as well as a description of observer rights as applicable. *This list may be submitted as an attachment.*

10. Does your entity, whether as borrower, surety, guarantor, or other, have any indebtedness, liabilities, or obligations to an entity based in, or funded by, any foreign country of concern? ☐ Yes ☐ No

    If yes, disclose the name of the entity and the details of the obligation.

11. *Will the project identified in Section 1 include any of the following? ☐ Yes ☒ No
    - Federal Acquisition Security Council (FASC)-prohibited unmanned aircraft systems or the operation thereof.
    - Coded equipment where the source code is written in a foreign country of risk.
    - Equipment from a foreign country of concern that will be connected to the internet or other remote communication system.
    - Equipment used on the project after delivery where any entity or individual based in a foreign country of concern will have physical or remote access to any part of the equipment.

    If yes, disclose the name and address of the supplier, and the item proposed for use on the project:

12. * Is the DOE funding for the project identified in Section 1 supporting the deployment of battery energy storage systems (BESS)? ☐ Yes ☒ No

    If yes, please answer the following questions:

    a. Please identify the vendor/integrator of any battery energy storage systems (BESS) covered under the award. Provide the model name/number of the BESS if possible.

b. Please identify the inverter manufacturer for the BESS covered under the award. If the inverter manufacturer is the same entity as the primary BESS vendor/integrator, please provide that confirmation.

c. Please identify the supplier of the battery management systems (BMS) for the BESS covered under the award. If the BMS provider is the same entity as the primary BESS vendor/integrator, please provide that confirmation.

d. Please identify the manufacturer of the battery packs included in the BESS covered under the award. If the battery pack provider is the same entity as the primary BESS vendor/integrator, please provide that confirmation.

e. Please identify the manufacturer of the battery cells included in the BESS covered under the award. If the battery cell provider is the same entity as the primary BESS vendor/integrator, please provide that confirmation.

Section 3. Certifications

Individuals participating in a Malign Foreign Talent Recruitment Program are prohibited from participating in a DOE award. The recipient and the subrecipients must certify that 1) the covered individuals in their respective employment have been made aware of the Malign Foreign Talent Recruitment Program prohibition and 2) the individuals have complied with their responsibilities to annually certify they are not party to a Malign Foreign Talent Recruitment Program.

<table>
<tr><td colspan="2">I am providing this disclosure and certification on behalf of the above-mentioned entity as part of my responsibilities as the entity's authorized official for the above-mentioned project. I certify to the best of my knowledge and belief that the information contained in this document is true, complete, and accurate.

I further certify that 1) the covered individuals in their respective employment have been made aware of the Malign Foreign Talent Recruitment Program prohibition and 2) the covered individuals have complied with their responsibilities to certify they are not party to a Malign Foreign Talent Recruitment Program.

I understand any false fictitious, or fraudulent information, misrepresentations, half-truths, or omissions of material fact, may subject me to criminal, civil, or administrative penalties for fraud, false statements, false claims or otherwise. (18 U.S.C. §§ 1001 and 287, and 31 U.S.C. §§ 3729, 3733, and 3801-3812). I further understand and agree that (1) the statements and representations made in this document are material to DOE's funding decision; (2) in the event DOE issues an award, should I later learn during the life of the award that an individual participating in the project funded under the award is or is reasonably believed to be involved with a Malign Foreign Talent Recruitment Program, I must notify the cognizant DOE/NNSA Grants Officer within five business days on learning such information; and (3) I have an ongoing responsibility to update the disclosures provided above during the period of performance of the award if there is any change to a disclosure.</td></tr>
<tr><td>Authorized Official:
Raquel Andersen-Hubby on behalf of Robin B. Riglin</td><td>Title:
Senior Proposal Generalist</td></tr>
<tr><td>Signature:
Raquel Andersen-Hubby</td><td>Date:
April 17, 2026</td></tr>
</table>

Appendix A. Definitions

1. Covered Individual. The Department of Energy (DOE) has designated individuals who contribute in a substantive, meaningful way to the project proposed to be carried out with an award from DOE, at both the prime and subrecipient levels. Often, these individuals have doctoral or other professional degrees, although individuals at the master's or PhD level may be considered covered individuals if their involvement meets this definition. Consultants, graduate students, and those with a postdoctoral role also may be considered covered individuals if they meet this definition. Consult the Notice of Funding Opportunity or Award Terms and Conditions, as applicable, for the relevant definition of "covered individual."

2. Foreign Country of Concern. The term "foreign country of concern" means the People's Republic of China, the Democratic People's Republic of Korea, the Russian Federation, the Islamic Republic of Iran, or any other country determined to be a country of concern by the Secretary of State, and including countries of risk designated by the Department.

3. Malign Foreign Talent Recruitment Program. As defined in P.L. 117-167, Section 10638 (4), the term "malign foreign talent recruitment program" means:

(A) any program, position, or activity that includes compensation in the form of cash, in-kind compensation, including research funding, promised future compensation, complimentary foreign travel, things of non de minimis value, honorific titles, career advancement opportunities, or other types of remuneration or consideration directly provided by a foreign country at any level (national, provincial, or local) or their designee, or an entity based in, funded by, or affiliated with a foreign country, whether or not directly sponsored by the foreign country, to the targeted individual, whether directly or indirectly stated in the arrangement, contract, or other documentation at issue, in exchange for the individual –

(i) engaging in the unauthorized transfer of intellectual property, materials, data products, or other nonpublic information owned by a United States entity or developed with a Federal research and development award to the government of a foreign country or an entity based in, funded by, or affiliated with a foreign country regardless of whether that government or entity provided support for the development of the intellectual property, materials, or data products;
(ii) being required to recruit trainees or researchers to enroll in such program, position, or activity;
(iii) establishing a laboratory or company, accepting a faculty position, or undertaking any other employment or appointment in a foreign country or with an entity based in, funded by, or affiliated with a foreign country if such activities are in violation of the standard terms and conditions of a Federal research and development award;
(iv) being unable to terminate the foreign talent recruitment program contract or agreement except in extraordinary circumstances;
(v) through funding or effort related to the foreign talent recruitment program, being limited in the capacity to carry out a research and development award or required to engage in work that would result in substantial overlap or duplication with a Federal research and development award;
(vi) being required to apply for and successfully receive funding from the sponsoring foreign government's funding agencies with the sponsoring foreign organization as the recipient;

(vii) being required to omit acknowledgment of the recipient institution with which the individual is affiliated, or the Federal research agency sponsoring the research and development award, contrary to the institutional policies or standard terms and conditions of the Federal research and development award;

(viii) being required to not disclose to the Federal research agency or employing institution the participation of such individual in such program, position, or activity; or

(ix) having a conflict of interest or conflict of commitment contrary to the standard terms and conditions of the Federal research and development award; and

(B) a program that is sponsored by—

(i) a foreign country of concern or an entity based in a foreign country of concern, whether or not directly sponsored by the foreign country of concern.

(ii) an academic institution on the list developed under section 1286(c)(8) of the John S. McCain National Defense Authorization Act for Fiscal Year 2019 (10 U.S.C. 2358 note; 1 Public Law 115–232); or

(iii) a foreign talent recruitment program on the list developed under section 1286(c)(9) of the John S. McCain National Defense Authorization Act for Fiscal Year 2019 (10 U.S.C. 2358 note; 1 Public Law 115–232).

Appendix B. Identification of Confidential Information

If the response includes business sensitive information (e.g., commercial or financial information that is privileged or confidential), trade secrets, proprietary, or otherwise confidential information in their or otherwise confidential information, it is furnished to the federal government (government) in confidence and such information will be withheld from public disclosure to the extent permitted by law, including the Freedom of Information Act. In accordance with 10 CFR part 1004, DOE will make its own determination about the confidential status of the information and treat it according to its determination.

The response must be marked "confidential" and clearly and conspicuously identify the business sensitive, trade secrets, proprietary, or otherwise confidential information in accordance with the instructions below. Failure to comply with these marking requirements may result in the disclosure of the unmarked information under the Freedom of Information Act or otherwise. The government is not liable for the disclosure or use of unmarked information and may use or disclose such information for any purpose as authorized by law.

The first page of the response must be marked as follows and identify the specific pages business sensitive, trade secrets, proprietary, or otherwise confidential information:

> **Notice of Restriction on Disclosure and Use of Data**:
> Pages [list applicable pages] of this document may contain business sensitive, trade secrets, proprietary, or otherwise confidential information that is exempt from public disclosure. Such information shall be used or disclosed only for evaluation purposes or in accordance with a financial assistance agreement between the submitter and the government. The government may use or disclose any information that is not appropriately marked or otherwise restricted, regardless of source. [End of Notice]

In addition, (1) the header and footer of every page that contains business sensitive, trade secrets, proprietary, or otherwise confidential information must be marked as follows: "Contains Business Sensitive, Trade Secrets, Proprietary, or Otherwise Confidential Information Exempt from Public Disclosure," and (2) every line or paragraph containing such information must be clearly marked with double brackets or highlighting. DOE will make its own determination about the confidential status of the information and treat it according to its determination.